Practice workbook · Draw a bar chart and justify its use

Data for separate categories or grouped classes are shown as bars; may include adding an axis scale or saying why bars suit the data.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-BAR-CHART

1. Question 5 · part b.i · 5 marks

January 2022 Paper 1

Worked-example retry · Attempt part b.i only.

5 (a) Describe the possible effects of a DNA mutation.

(3)

(b) Changes to whole chromosomes can happen when meiosis does not occur properly.

These changes are called chromosome aberrations.

The table shows the probability of children being born with a chromosome aberration to mothers of different ages.

Age of mother in years	15–19	20–24	25–29	30–34	35–39	40–44
Probability of chromosome aberration	0.0010	0.0010	0.0010	0.0015	0.0045	0.0140

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1. Question 5 · part b.i · 5 marks

January 2022 Paper 1

Worked-example retry · Attempt part b.i only.

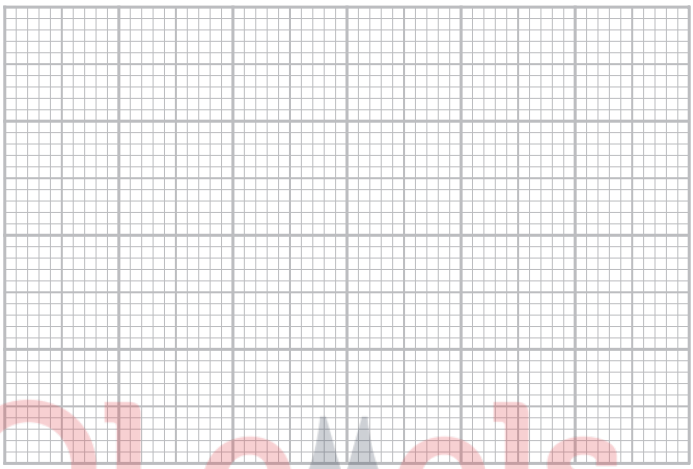
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(i) Plot a bar chart of the results.

(5)



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(ii) In one year, 43 000 children are born to mothers aged 30–34.

Calculate the probable number of these children who will have a chromosome aberration.

Give your answer to two significant figures.

(3)

number of children = ...

P 6 7 0 5 9 A 0 1 5 2 4

Turn over ▶

1. Question 5 · part b.i · 5 marks · Working and answer

Working and answer

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2. Question 4 · part b.i · 3 marks

November 2020 Paper 1R

Further practice · Attempt part b.i only.

4 (a) Kwashiorkor is a deficiency disease that results from the lack of a certain nutrient in the diet.

Use words from the box to complete the passage about kwashiorkor.

(2)

fingers	carbohydrate	bread	ankles	protein	fat

Kwashiorkor is a disease that is more common in children.

One symptom of kwashiorkor is swelling in the ..... .

The symptoms of kwashiorkor are corrected by adding more ..... to the diet.

(b) The table gives information about the number of cases of kwashiorkor recorded in a hospital over four years.

Year	Number of cases of kwashiorkor
2010	32
2011	21
2012	23
2013	19

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2. Question 4 · part b.i · 3 marks

November 2020 Paper 1R

Further practice · Attempt part b.i only.

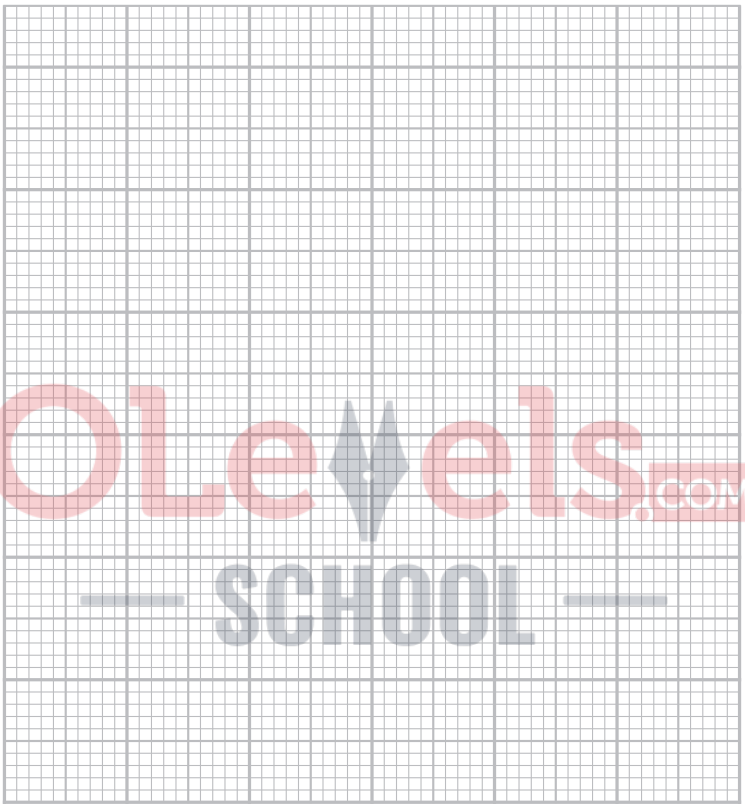
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(i) Plot a bar chart of the data on the grid.

(3)



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(ii) Suggest a reason for the difference in the number of recorded cases of kwashiorkor in 2010 compared with 2013.

(1)

.....

.....

(iii) In a hospital, 95 out of a total of 1978 children with malnutrition were found to have kwashiorkor.

Calculate the percentage of children with kwashiorkor.

(2)

percentage of children = ... %

(Total for Question 4 = 8 marks)

2. Question 4 · part b.i · 3 marks · Working and answer

Working and answer

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3. Question 3 · part a.i · 5 marks

November 2020 Paper 1

Further practice · Attempt part a.i only.

3 There are many factors that affect the height of a human.

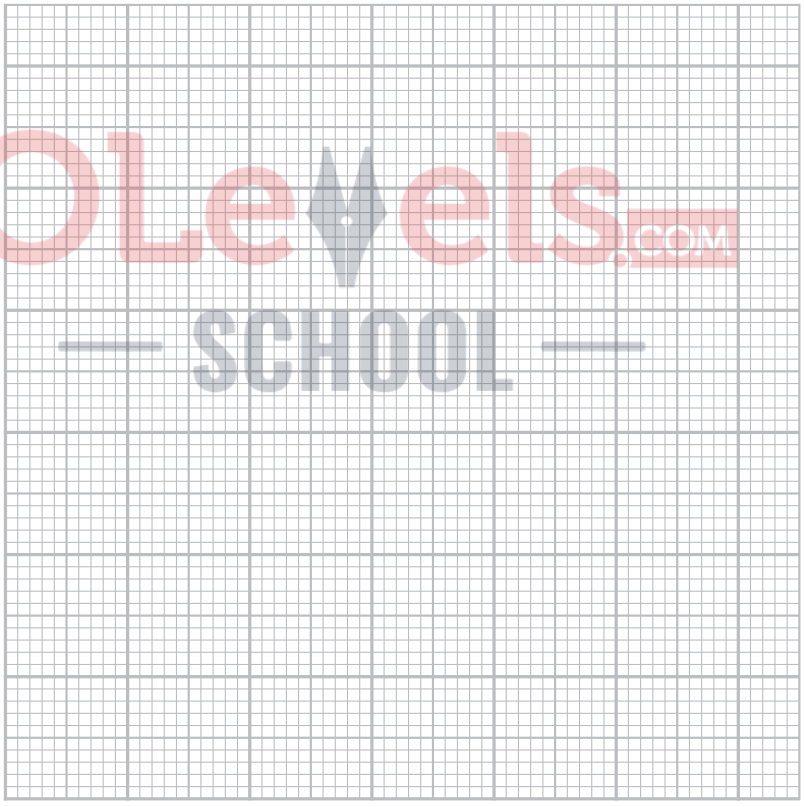
(a) The heights of a group of students are measured.

The table shows the results.

Height in cm	150–151	152–153	154–155	156–157	158–159	> 160
Number of students	4	14	25	20	10	4

(i) Plot a bar chart of the results.

(5)



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3. Question 3 · part a.i · 5 marks

November 2020 Paper 1

Further practice · Attempt part a.i only.

(ii) Describe the trend shown in this bar chart.

(2)

(b) State three factors that could affect the heights of the students.

(3)

1 .....

2 .....

3 .....

(Total for Question 3 = 10 marks)

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3. Question 3 · part a.i · 5 marks · Working and answer

Working and answer

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4. Question 4 · part a.i · 4 marks

November 2021 Paper 2

Further practice · Attempt part a.i only.

4 The table shows the results of a survey about the number of teeth per student in a class of young students.

Number of teeth	Number of students
23	1
24	2
25	3
26	4
27	7
28	13

(a) (i) Draw a bar chart to show the results of the survey.

(4)



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4. Question 4 · part a.i · 4 marks

November 2021 Paper 2

Further practice · Attempt part a.i only.

(ii) A full set of teeth for the students is 28

Calculate the percentage of the students in the class who have a full set of teeth.

(3)

percentage = ... %

(b) A full set of teeth for an adult is 32

Suggest why the full set of teeth for these students is only 28

(2)

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4. Question 4 · part a.i · 4 marks · Working and answer

Working and answer

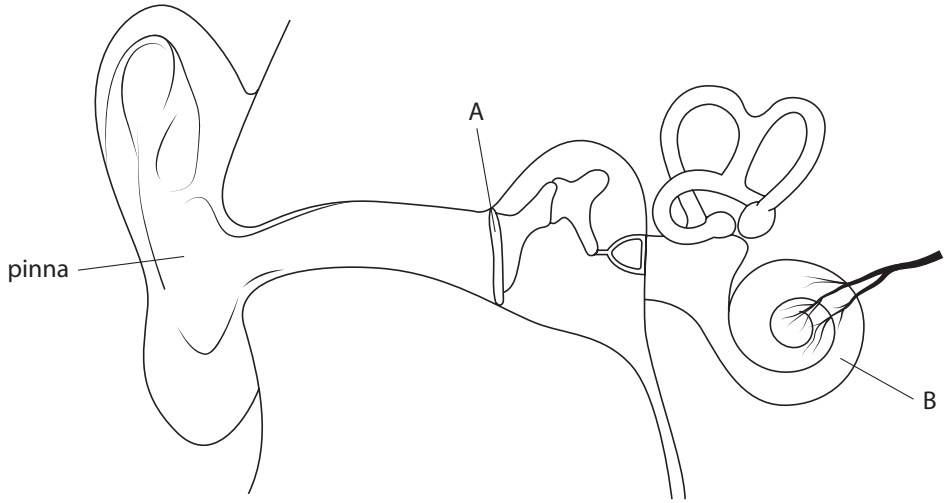
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5. Question 5 · part b.i · 4 marks

June 2022 Paper 2

Further practice · Attempt part b.i only.

5 The diagram shows a section through the human ear.



(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

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5. Question 5 · part b.i · 4 marks

June 2022 Paper 2

Further practice · Attempt part b.i only.

(b) The range of frequencies audible to the human ear can vary with age.

The table shows the maximum frequency detected by people of different ages.

Age in years	Maximum frequency detected in kHz
20	20
30	18
40	16
50	14

(i) Draw a bar chart to show how the maximum frequency detected varies with age.

(4)

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5. Question 5 · part b.i · 4 marks

June 2022 Paper 2

Further practice · Attempt part b.i only.

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(ii) Calculate the percentage change in the maximum frequency detected by people aged 20 and people aged 50.

(3)

percentage change = ... %

(c) A student suggests this method that could be used to produce the data shown in the table.

- select a person of each age to be tested
- use a signal generator to produce sounds at different frequencies
- record the maximum frequency detected by each person

(i) Give two reasons why this is unlikely to produce valid results.

(2)

1 .....

.....

2 .....

.....

(ii) Describe how the reliability of the results could be improved.

(2)

.....

.....

.....

.....

(Total for Question 5 = 15 marks)

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5. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

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6. Question 2 · part a.i · 4 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

2 (a) The table gives information about different methods of food preservation.

It also gives information on the length of time in days that each method keeps foods fresh.

Method of food preservation	Time food stays fresh in days
refrigeration	5
freezing	90
pasteurisation	35
vacuum-packed food	14

(i) Draw a bar chart to show the information in the table.

(4)

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P 7 4 6 2 6 R A 0 6 2 4

6. Question 2 · part a.i · 4 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

(ii) State how many of the methods shown in the bar chart use heat treatment to preserve food.

(1)

.....

(b) Bacteria can multiply rapidly in food that is not preserved.

Use words from the box to complete the passage about the growth of bacteria.

(6)

exponential	more than	stationary	equal to	lag
dying	death	less than	live	growth

The first stage in the growth of bacterial colonies is known as the ..... phase. During this phase, bacteria are not dividing.

The second stage of bacterial growth is known as the ..... phase. During this phase the bacteria are multiplying rapidly.

In the third stage, nutrients are running out and the number of dying cells is ..... the number of dividing cells.

Finally, during the ..... phase the number of dividing cells is ..... the number of ..... cells.

(Total for Question 2 = 11 marks)

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6. Question 2 · part a.i · 4 marks · Working and answer

Working and answer

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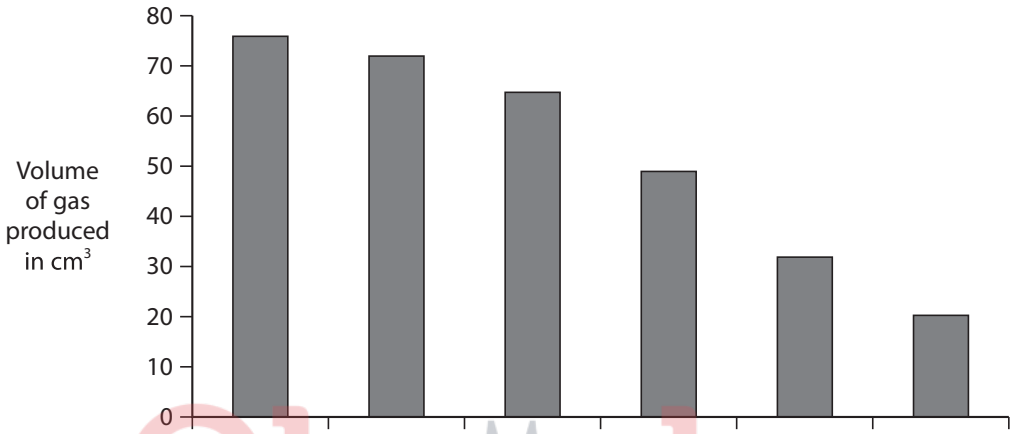
7. Question 3 · part a.i · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.i only.

3 A student investigates a reaction. The reaction is catalysed by an enzyme and produces a gas. The student collects the gas for 10 minutes and measures the volume of the gas. The student repeats this every 10 minutes for one hour.

The bar chart shows the student's results.



(a) (i) Add a suitable labelled scale to the *x*-axis.

(2)

(ii) Estimate the total volume of gas produced after 30 minutes.

(2)

volume of gas = ... cm^3

(iii) Describe the trend shown by the results.

(3)

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7. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

3 (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

(4)

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(ii) Describe the function of a motor neurone.

(2)

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8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2R

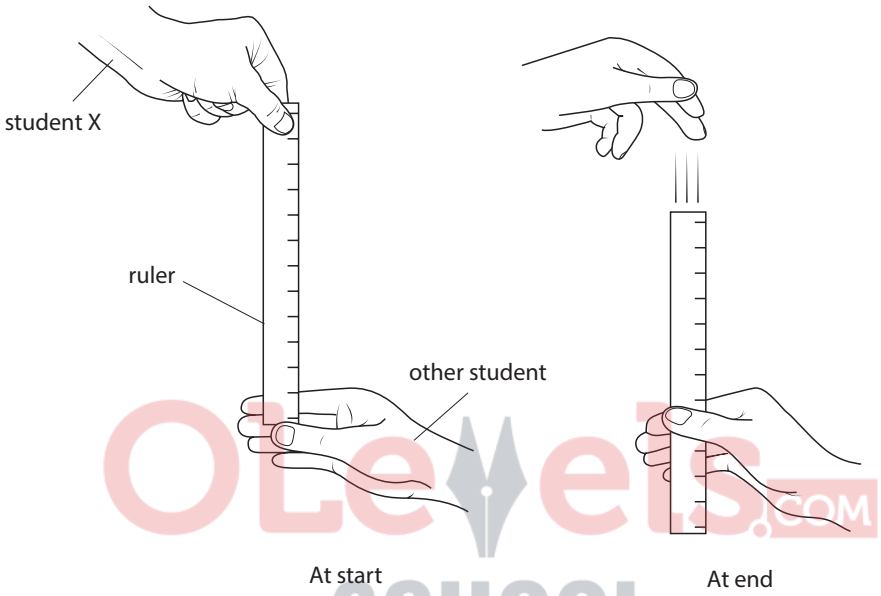
Further practice · Attempt part b.iii only.

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(b) Student X uses this equipment to investigate the reaction time of another student.



Student X knows that the reaction time can be calculated using this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

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8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

(i) Describe a method the student could use to compare the reaction times of 10 students.

(6)

(ii) Name the dependent variable in this investigation.

(1)

(iii) Give a reason why the student should display the results using a bar chart instead of a line graph.

(1)

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8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

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(iv) The reading on the ruler for one of the students is 0.15 m.

Calculate the reaction time of this student.

Use this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

(2)

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reaction time = ... s

(Total for Question 3 = 16 marks)

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Working and answer